

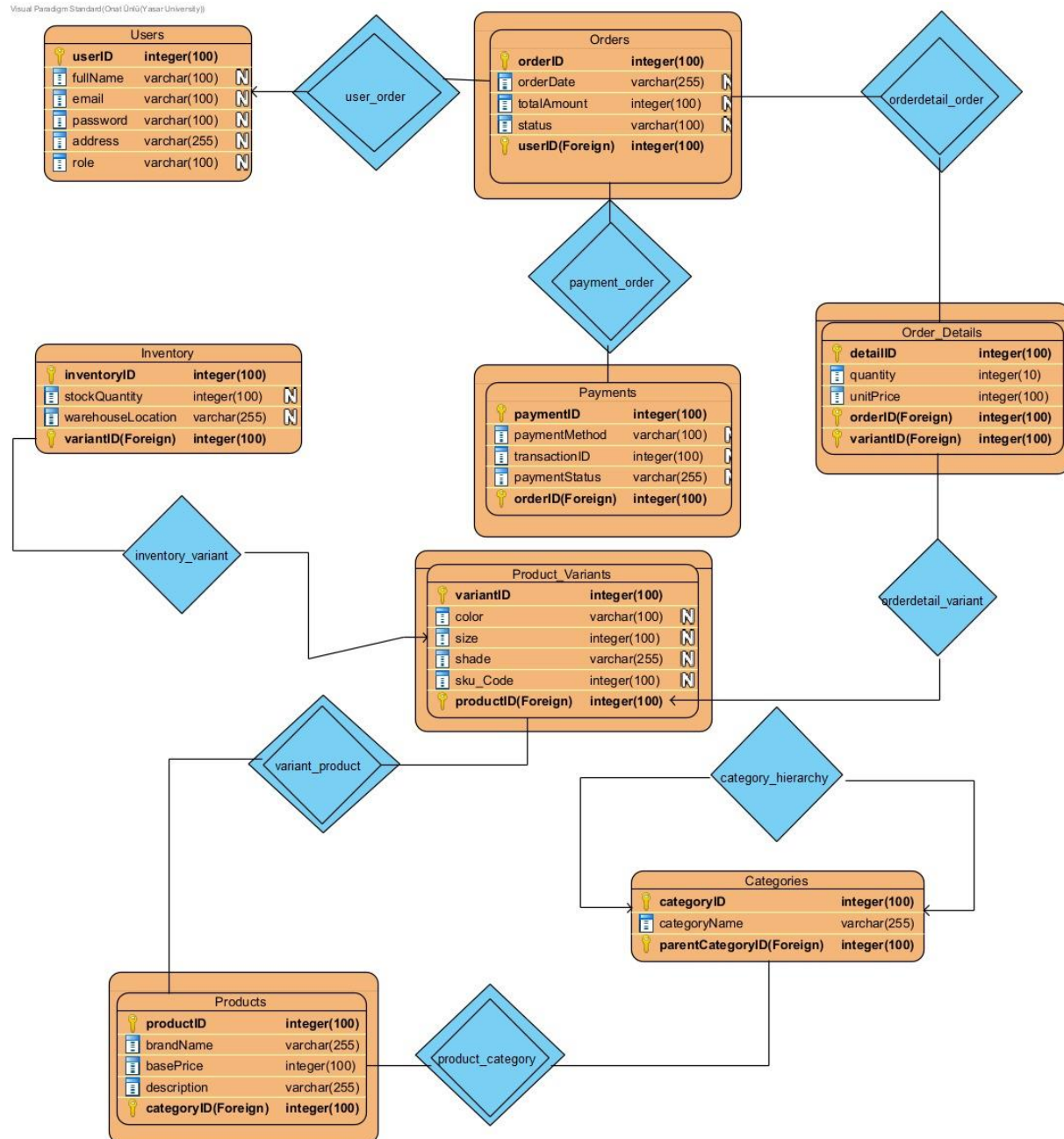
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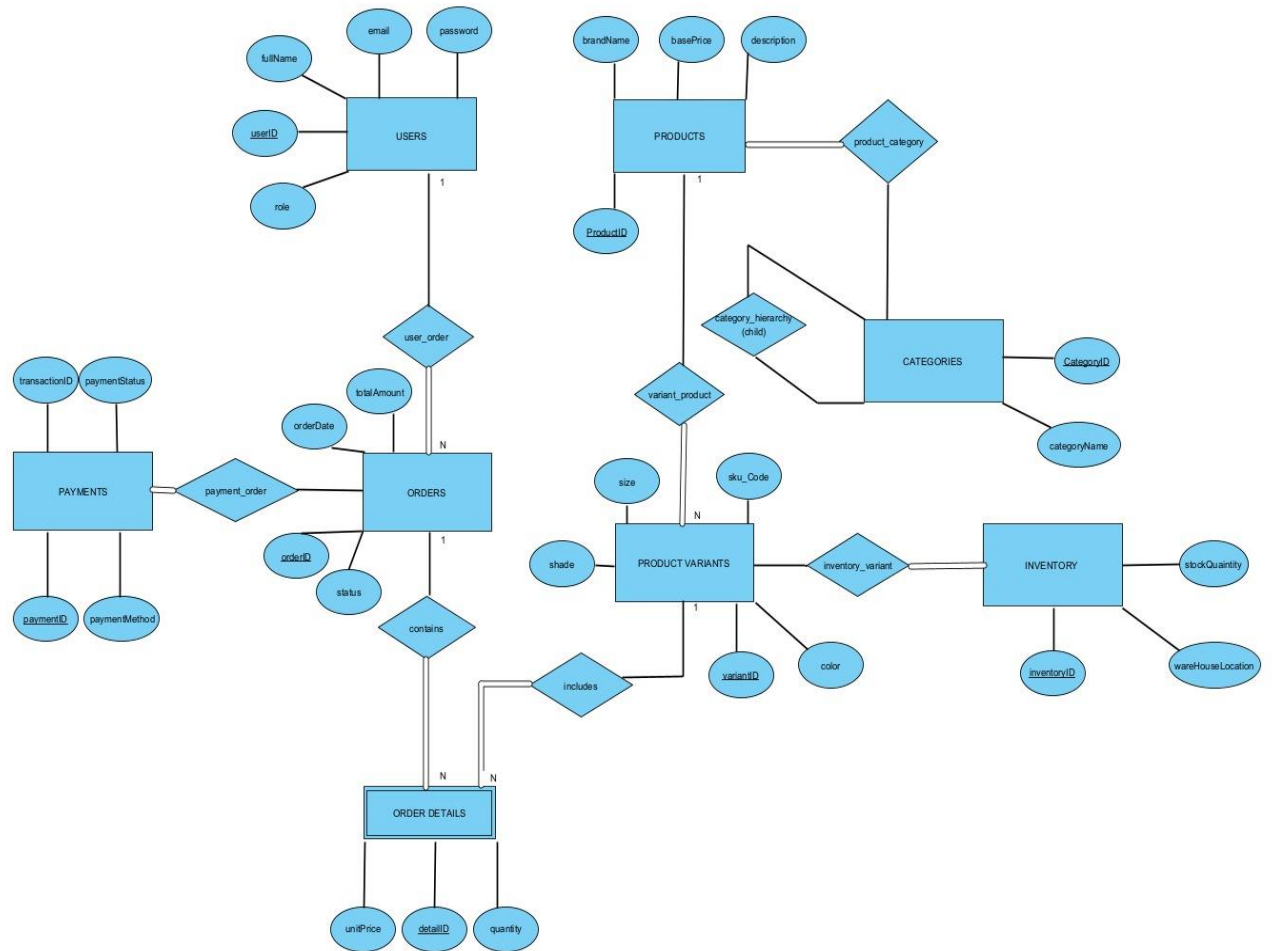
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ER DIAGRAM





Reduction to Relation Schema

Entities

- **users**(userID, fullName, email, password, address, role)
- **categories**(categoryID, categoryName)
- **products**(productID, brandName, basePrice, description)
- **product_variants**(variantID, color, size, shade, sku_Code)
- **inventory**(inventoryID, stockQuantity, warehouseLocation)
- **orders**(orderID, orderDate, totalAmount, status)
- **order_details**(detailID, quantity, unitPrice)
- **payments**(paymentID, paymentMethod, transactionID, paymentStatus)

Relationships

- **user_order**(userID, orderID)
- **category_hierarchy**(categoryID, parentCategoryID)
- **product_category**(productID, categoryID)
- **variant_product**(variantID, productID)
- **inventory_variant**(inventoryID, variantID)
- **order_user**(orderID, userID)
- **orderdetail_order**(detailID, orderID)
- **orderdetail_variant**(detailID, variantID)
- **payment_order**(paymentID, orderID)

Final Relation Schema

1. Users

- **users** (userID, fullName, email, password, address, role)

2. Categories

- **categories** (categoryID, categoryName, parentCategoryID)
 - parentCategoryID → categories(categoryID)

3. Products

- **products** (productID, categoryID, brandName, basePrice, description)
 - categoryID → categories(categoryID)

4. Product Variants

- **product_variants** (variantID, productID, color, size, shade, sku_Code)
 - productID → products(productID)

5. Inventory

- **inventory** (inventoryID, variantID, stockQuantity, warehouseLocation)
 - variantID → product_variants(variantID)

6. Orders

- **orders** (orderID, UserID, orderDate, totalAmount, status)
 - userID → users(userID)

7. Order Details

- **order_details** (detailID, orderID, variantID, quantity, unitPrice)
 - orderID → orders(orderID)
 - variantID → product_variants(variantID)

8. Payments

- **payments** (paymentID, orderID, paymentMethod, transactionID, paymentStatus)
 - orderID → orders(orderID)

